

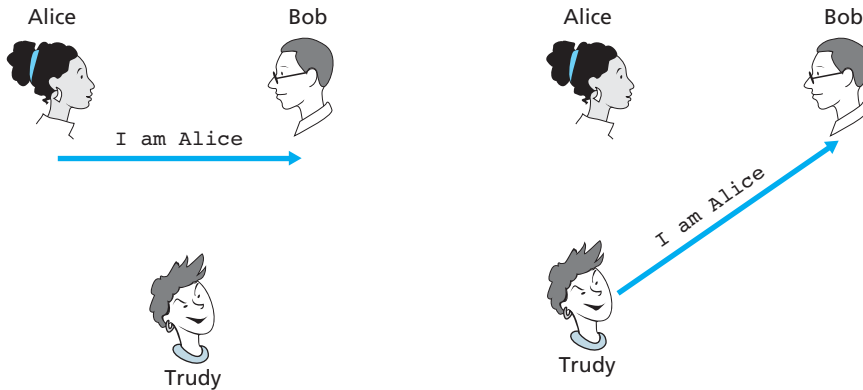


Figure 8.15 ♦ Protocol *ap1.0* and a failure scenario

that the person sending the message “I am Alice” is indeed Alice. For example, Trudy (the intruder) could just as well send such a message.

8.4.2 Authentication Protocol *ap2.0*

If Alice has a well-known network address (e.g., an IP address) from which she always communicates, Bob could attempt to authenticate Alice by verifying that the source address on the IP datagram carrying the authentication message matches Alice’s well-known address. In this case, Alice would be authenticated. This might stop a very network-naïve intruder from impersonating Alice, but it wouldn’t stop the determined student studying this book, or many others!

From our study of the network and data link layers, we know that it is not that hard (for example, if one had access to the operating system code and could build one’s own operating system kernel, as is the case with Linux and several other freely available operating systems) to create an IP datagram, put whatever IP source address we want (for example, Alice’s well-known IP address) into the IP datagram, and send the datagram over the link-layer protocol to the first-hop router. From then on, the incorrectly source-addressed datagram would be dutifully forwarded to Bob. This approach, shown in Figure 8.16, is a form of IP spoofing. IP spoofing can be avoided if Trudy’s first-hop router is configured to forward only datagrams containing Trudy’s IP source address [RFC 2827]. However, this capability is not universally deployed or enforced. Bob would thus be foolish to assume that Trudy’s network manager (who might be Trudy herself) had configured Trudy’s first-hop router to forward only appropriately addressed datagrams.

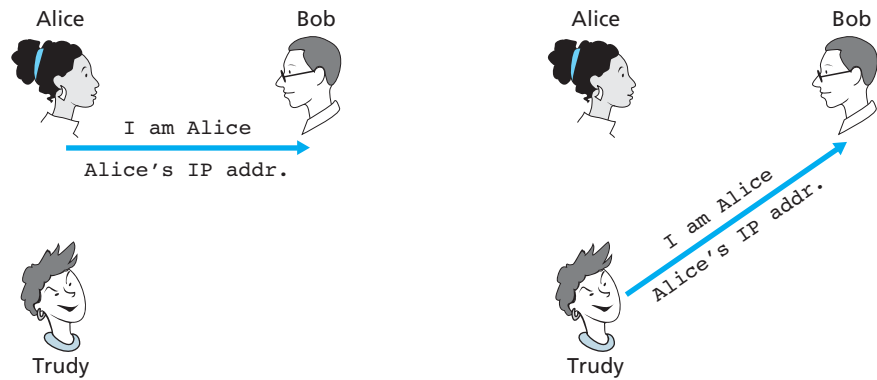


Figure 8.16 ♦ Protocol *ap2.0* and a failure scenario

8.4.3 Authentication Protocol *ap3.0*

One classic approach to authentication is to use a secret password. The password is a shared secret between the authenticator and the person being authenticated. Gmail, Facebook, telnet, FTP, and many other services use password authentication. In protocol *ap3.0*, Alice thus sends her secret password to Bob, as shown in Figure 8.17.

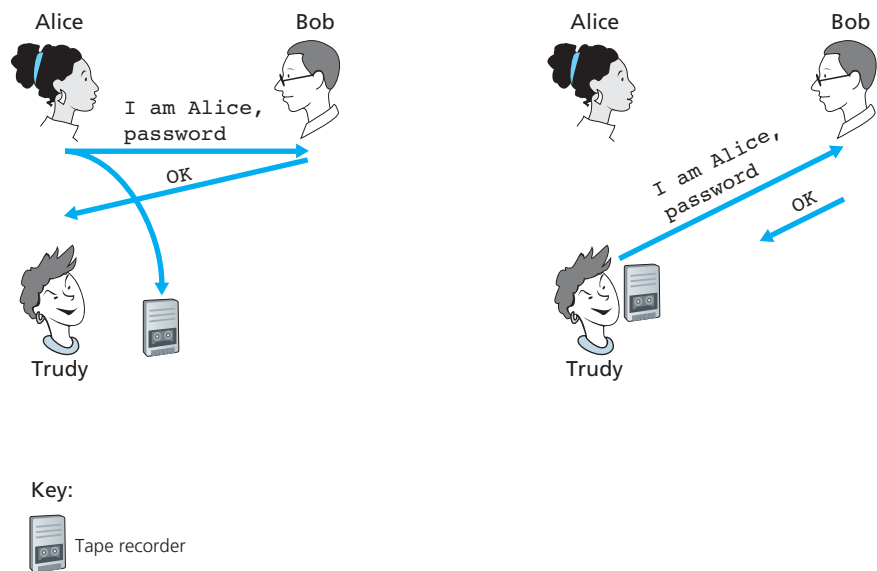


Figure 8.17 ♦ Protocol *ap3.0* and a failure scenario